# Budget Anchored Region Reduction for Collaborative Responsibility Repair

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***Abstract*—Localized cost changes can require responsibility transfers under a hard handoff budget. We derive Budget Anchored Region Reduction (BARR), a safe exterior-fixing rule from perturbation support, positive handoff costs and a certified old-plan gap $\varepsilon$. Reverting all unanchored modification components together incurs this gap once. The restricted optimum preserves the full budgeted value at $\varepsilon = 0$ and loses at most $\varepsilon$ otherwise; at least one suitable solution is retained. Boundary absorption transfers restricted solver bounds to the full problem. Matched CPU experiments show conditional benefits: on 512-vertex modular graphs, median paired speed is 3.17 on queries not resolved by the shared cut shortcut but 1.01 over all queries. Dense graphs can retain the whole model, and hard cores can remain unresolved.**

***Index Terms*—collaborative design, responsibility allocation, budgeted repair, region reduction, optimality certificate**

## I. Introduction

Local cost changes motivate collaborative responsibility transfers, while documentation, onboarding and interface handoffs impose a hard change budget.

We study primary maintenance-contact selection between two eligible teams; unary costs price stewardship and pairwise costs price cross-team interfaces. GitHub permits alternative team owners and accepts any owner's approval for its code-owner requirement [1]. Fluent UI publicly lists the same two teams for several packages [2]. Eligibility declarations do not measure skill or spare capacity. Our static-slack check certifies capacity redundancy throughout the budget ball; active balance and scheduling remain outside scope.

Graph cuts minimize unconstrained binary attractive energies [3], but their returned optimum may exceed the change budget. Distance alone cannot justify fixing distant labels: a suboptimal old plan may have valuable repairs anywhere.

Our structural reduction uses changed unary factors as anchors. The old gap bounds the cost of reverting all unanchored modification components together. Remaining modifications have budget-feasible paths to anchors, yielding an activity region and

$$\mathrm{OPT}_B \le \mathrm{OPT}_K \le \mathrm{OPT}_B + \varepsilon, \tag{1}$$

where $\mathrm{OPT}_B$ is the full budgeted optimum and $\mathrm{OPT}_K$ fixes exterior labels to the old plan. At $\varepsilon = 0$, at least one suitable optimum is preserved, not necessarily every tied optimum.

Ownership records ground the role; CAD co-edit graphs [4], [5] test structures with synthetic costs. Neither is an industrial trial.

The contribution is one theoretical chain: exterior fixing with an additive old-gap certificate. Dijkstra, boundary absorption and CP-SAT implement it; matched experiments test conditional speed and certificate tightness. Code, sources, outcomes and supplementary analyses: https://github.com/elexlebron-spec/cscwd-11.

## II. Related Work

CSCWD allocation topics include personnel sharing [6], cooperation/conflict [7], and qualified, capacity-limited divisible workloads [8]. Our model addresses a narrower responsibility layer; it does not enforce or outperform those different resource models.

Table I identifies the needed information. A budget neighborhood bounds allowed moves but does not identify safely fixed exterior labels. Fixed-objective screening certifies that minimization; here the hard budget, perturbation support and old gap must be used jointly. The cited formulations do not directly state this guarantee. Their extensions may need separate analysis.

Dynamic cuts reuse residual flows for unconstrained updates [14]; stable dynamic cuts concern vertex arrival and flip stability [15]. They do not directly supply the stated fixed-graph budget certificate. Nor can unconstrained persistency simply be reused: with isolated vertices, $x^0 = (0, 0)$, $B = 1$, and new unary pairs $(0, -2), (0, -1)$, fixing both unconstrained-optimal labels to $(1, 1)$ removes every feasible repair. Section IV-E shows why the old certificate is also necessary. Neither counterexample rules out deriving our result from some other prior theorem.

## III. Problem Formulation

Let $G = (V, E)$ be fixed and undirected, with $n = |V|$ and $m = |E|$. Vertex $i$ is a work package and $x_i \in \{0, 1\}$ selects a primary maintenance contact from the same two prequalified teams throughout $G$. Node-specific candidate pairs would invalidate the cross-team meaning of label inequality. For $t \in \{0, 1\}$, define

$$E_t(x) = \sum_{i \in V} c_i^t(x_i) + \sum_{ij \in E} w_{ij} \mathbf{1}[x_i \neq x_j], \tag{2}$$

with integer unary costs and fixed integer $w_{ij} \ge 0$. Given old assignment $x^0$, handoff effort is

$$R(x, x^0) = \sum_{i \in V} r_i \mathbf{1}[x_i \neq x_i^0], \qquad r_i \in \mathbb{Z}_{>0}. \tag{3}$$

TABLE I
CLOSEST FORMULATIONS. MISSING ASSUMPTIONS ARE NOT REQUIRED IN THE CITED FORMULATION, RATHER THAN IMPOSSIBLE EXTENSIONS.

| Work | Perturbation setting | Modification control | Old-plan requirement | Guarantee or objective |
|---|---|---|---|---|
| Local branching [9] | General binary model | Hamming neighborhood | Feasible incumbent | Explore an incumbent neighborhood by integer optimization |
| Binary-energy trust region [10] | Binary pairwise energy | Hamming bound / dual | Current iterate | Graph-cut Lagrangian bound in energy approximation |
| Submodular reassignment [11] | Multiple task scenarios | Distances to minimizer sets | First-stage plan is a variable | Optimize scenario-wise reassignment distances |
| OptiRefine [12] | Densest-subgraph / maximum-cut objectives | Exactly $k$ unit refinements | Given graph solution | Approximation, hardness, and refinement algorithms |
| Safe SFM screening [13] | Fixed submodular objective | No handoff-budget model | Proximal optimization certificate | Fix active/inactive elements for that minimization |
| BARR | Local unary changes on a fixed graph | Weighted upper budget $B$ | Certified old-objective gap | Fix outside $K_B$; exact value at zero gap, additive loss at most the gap |

For integer $B \geq 0$, the repair is

$$\mathrm{OPT}_B = \min\{E_1(x) : x \in \{0,1\}^V,\ R(x, x^0) \leq B\}. \quad (4)$$

The old plan is feasible; there are no additional hard constraints. Stewardship costs $c_i^1$ affect the objective, whereas handoff costs $r_i$ consume the change budget.

Only unary factors on $A = \{i : c_i^0 \neq c_i^1\}$ may change. Both positive and negative shifts are allowed. We require a certified $\varepsilon \geq 0$ satisfying

$$E_0(x^0) \leq \min_x E_0(x) + \varepsilon. \quad (5)$$

Our implementation computes the old unconstrained minimum exactly by graph cut, making $\varepsilon$ the actual old-objective gap. A valid old lower bound can instead supply a conservative certificate. Optimality cannot be inferred from an organization's use of the old plan. Negative unary entries are handled by retaining label-independent offsets when constructing nonnegative terminal capacities.

## IV. BUDGET ANCHORED REGION REDUCTION

### A. Reverting Remote Modification Components

For a budget-feasible $x$, let $S = \{i : x_i \neq x_i^0\}$. A connected component of the induced graph $G[S]$ is anchored if it intersects $A$.

*Lemma 1 (Union reversion):* Let $T$ be the union of all components of $G[S]$ disjoint from $A$. Restore $x_i^0$ on $T$ and retain $x_i$ elsewhere, obtaining $y$. Then $R(y, x^0) \leq R(x, x^0)$ and $E_1(y) \leq E_1(x) + \varepsilon$.

*Proof:* Let $z$ equal $x$ on $T$ and $x^0$ elsewhere. No edge joins $T$ to $S \setminus T$, since these sets contain different components of $G[S]$. Every neighbor outside $S$ keeps its old label in both $x$ and $y$. Moreover, $c_i^1 = c_i^0$ on $T$. Expanding unary and crossing-edge differences gives

$$E_1(x) - E_1(y) = E_0(z) - E_0(x^0) \geq -\varepsilon. \quad (6)$$

The inequality follows from (5). Restoring vertices removes positive resource terms. Applying the old-gap bound once to the union $T$ incurs one $\varepsilon$, rather than one per component. ∎

Union reversion uses fixed interactions, separation of changed components and a valid old gap. Nonnegative weights enable our exact binary graph-cut implementation.

### B. A Budget-Defined Activity Region

Charge every vertex on a path $P : a \rightsquigarrow v$, including both endpoints. Define

$$d_r(A, v) = \min_{a \in A,\, P: a \rightsquigarrow v} \sum_{i \in P} r_i, \quad K_B = \{v : d_r(A, v) \leq B\}. \quad (7)$$

An unreachable vertex has infinite distance; $A = \varnothing$ gives $K_B = \varnothing$. Positive costs bound the path length of feasible changed components. An anchor with $r_a > B$ cannot itself be changed.

*Theorem 1 (Region reduction):* Under Section III's assumptions, let $\mathrm{OPT}_K$ be the optimum of (4) with $x_i = x_i^0$ outside $K_B$. Then (1) holds. If $\varepsilon = 0$, at least one global budgeted optimum is supported in $K_B$.

*Proof:* Take an optimum $x^*$ of (4) and apply Lemma 1 to obtain $y$. Every remaining changed component intersects $A$. For a changed $v$, a simple path to an anchor lies inside its component. Every path vertex is changed, so its node-cost sum is at most $R(y, x^0) \leq B$. Thus $v \in K_B$, and $y$ is feasible for the restricted problem. Hence $\mathrm{OPT}_K \leq E_1(y) \leq \mathrm{OPT}_B + \varepsilon$. Restricting the feasible set gives $\mathrm{OPT}_B \leq \mathrm{OPT}_K$. For $\varepsilon = 0$, the values coincide. ∎

For unit costs, $K_B$ is the at-most-$(B-1)$-hop neighborhood of $A$ when $B \geq 1$; $K_0 = \varnothing$. Charging the anchor is essential. Letting $r_{\min} = \min_i r_i$, define the coarse region

$$H_B = \{v : 1 + \mathrm{dist}_G(A, v) \leq \lfloor B / r_{\min} \rfloor\}. \quad (8)$$

A path's cost is at least $r_{\min}$ times its number of vertices, implying $K_B \subseteq H_B$. Fixing the exterior of $H_B$ is also safe, but can retain unnecessary vertices under heterogeneous costs. We use it as a mechanism ablation, not a separate contribution.

On path $a[3] - b[2] - c[1]$ with node costs in brackets, $A = \{a\}$ and $B = 5$ give distances $3, 5, 6$ and $K_B = \{a, b\}$.

### C. Boundary Absorption and Certificates

For each crossing edge $ij$, $i \in K = K_B$, $j \notin K$, absorb its interaction into the retained unary:

$$\widetilde{c}_i(a) = c_i^1(a) + \sum_{j \notin K:\, ij \in E} w_{ij} \mathbf{1}[a \neq x_j^0]. \quad (9)$$

Let $C$ sum exterior unaries at $x^0$ and edges entirely outside $K$. Retaining internal edges yields

$$E_1(x) = E_K(x_K) + C \tag{10}$$

for every reconstruction. Dropping crossing edges would solve a different problem. Only retained vertices consume the restricted change budget.

*Corollary 1 (Global interval):* If a restricted solver returns $L_K \le \mathrm{OPT}_K \le U_K$, including $C$, its feasible solution reconstructs to a feasible full assignment and

$$L_K - \varepsilon \le \mathrm{OPT}_B \le U_K. \tag{11}$$

*Proof:* Reconstruction preserves resource usage and, by (10), energy. Feasibility supplies the upper bound, and Theorem 1 gives $\mathrm{OPT}_B \ge \mathrm{OPT}_K - \varepsilon \ge L_K - \varepsilon$. ∎

*Corollary 2 (Merged lower bound):* Let $v = \min_x E_1(x)$. With $L_K, U_K$ including $C$, return $[L_{\mathrm{final}}, U_K]$, where

$$L_{\mathrm{final}} = \begin{cases} \max\{v, L_K\}, & K_B = V, \\ \max\{v, L_K - \varepsilon\}, & K_B \subsetneq V. \end{cases} \tag{12}$$

*Proof:* $v \le \mathrm{OPT}_B$; if $K_B = V$, $\mathrm{OPT}_K = \mathrm{OPT}_B$. Otherwise apply Corollary 1. ∎

Closed returned bounds certify global optimality; restricted optimality alone need not.

### D. Algorithm and Cost

Algorithm 1 connects the proof to the query pipeline. A budget-feasible new cut proves optimality and ends both matched pipelines. Otherwise BARR pays for the old certificate, region and boundary construction. Optional capacity checking precedes the cut and terminates on failure. It is disabled in base benchmarks and charged when enabled.

**Algorithm 1: Certified BARR query**

*Input:* $E_0, E_1, x^0, r, B$; fixed attractive graph, local unary changes, $r_i > 0$; optional loads/capacities.

1. If capacities supplied, compute $W_t(B)$; if any $C_t - L_t^0 < W_t(B)$, return `CAPACITY_NOT_CERTIFIED`.
2. $(v, \bar{x}) \leftarrow \mathrm{Cut}(E_1)$; if $R(\bar{x}, x^0) \le B$, return $\bar{x}, v, [v, v]$.
3. $m_0 \leftarrow \min_x E_0(x)$ by cut; $\varepsilon \leftarrow E_0(x^0) - m_0$.
4. $A \leftarrow \{i : c_i^0 \ne c_i^1\}$; node-weighted Dijkstra: $d(a) = r_a$, relax $d(j) \leftarrow \min\{d(j), d(i) + r_j\}$.
5. $K \leftarrow \{i : d(i) \le B\}$; fix exterior to $x^0$ and absorb crossing factors to obtain $(E_K, C)$.
6. $(x_K, [\ell, u]) \leftarrow \mathrm{BudgetSolve}(E_K, r_K, x_K^0, B)$, with valid lower bound $\ell$.
7. If no feasible incumbent at timeout, set $x_K \leftarrow x_K^0$, $u \leftarrow E_K(x_K^0)$; otherwise $u = E_K(x_K)$.
8. Reconstruct $x$; set $L_K = \ell + C$, $U_K = u + C$, $L_{\mathrm{final}}$ by (12). Return $x, U_K, [L_{\mathrm{final}}, U_K]$.

With an old certificate, support/region/boundary construction costs $O((n+m)\log n)$ time and $O(n+m)$ memory. Cold non-shortcut queries also pay for both cuts. With cut/solve costs $T_{\mathrm{cut}}, T_{\mathrm{solve}}$, total time is $T_{\mathrm{cut}}(E_1) + T_{\mathrm{cut}}(E_0) + O((n+m)\log n) + T_{\mathrm{solve}}(K, B)$. The backend can remain exponential; no polynomial-time budget algorithm or polynomial kernel follows. Low graph diameter can make $K_B = V$, leaving overhead without reducing the model.

Our CP-SAT formulation uses binary $x_i$ and edge variables $y_{ij} = |x_i - x_j|$, objective (2), and one linear budget constraint. Full and reduced graphs use the same formulation, settings, and old-plan hints.

### E. Why the Old Certificate Is Necessary

Take two disconnected vertices, $x^0 = (0, 0)$, $r = (1, 1)$, $B = 1$. Old unary pairs are $(0, 5)$ and $(10, 0)$; only the first changes to $(1, 5)$. The old gap is 10. The restricted optimum is 11; changing remote vertex two instead gives full optimum 1. The loss equals $\varepsilon$, disproving exact locality for arbitrary old plans.

### F. Checking Capacity Redundancy

For operator-supplied nonnegative integer loads $d_i^t$ and static capacities $C_t$, let $L_t^0$ be team $t$'s old load and compute

$$W_t(B) = \max_z \sum_i d_i^t z_i, \quad \sum_i r_i z_i \le B, \quad z_i \in \{0, 1\}, \quad z_i = 0 \text{ if } x_i^0 = t. \tag{13}$$

If $C_t - L_t^0 \ge W_t(B)$ for both teams, every budget-feasible repair respects capacity: incoming load is at most $W_t(B)$ and outflows only reduce load. Thus capacities are redundant and Theorem 1 still applies. A 0/1-knapsack dynamic program checks this in $O(nB)$ time; this optional cost is additional to BARR. This is an applicability check, not an active-capacity extension. Failure calls for a full capacity model. For example, two isolated unit-load vertices with $x^0 = (0, 1)$, $B = 2$, capacities $(1, 1)$, old costs $(0, 1), (1, 0)$ and new costs $(4, 0), (1, 0)$ have zero old gap. The capacity-feasible optimum swaps both (value 1), whereas fixing remote vertex two forces value 4. The check rejects this case. No scheduling guarantee follows from static slack.

## V. Experimental Evaluation

### A. Protocol and Earlier Mechanism Tests

Experiments use one arm64 MacBook Pro, Python 3.12.14, OR-Tools 9.15.6755 and one CP-SAT worker. FullCP/BARRCP share formulation, old-plan hint and new-cut shortcut. Timing includes cuts, preprocessing, model/solver construction, solving and reconstruction; BARR recomputes its old certificate. Common imports/warm-up and old-plan generation are excluded. Targets are soft. Inputs, configuration/source hashes, labels, bounds and all failures/incomplete solves are retained.

The main study has 480 combinations: $n = 128, 512$, three graph families, coupling 2/12, one/four anchors, $B = 8, 16$ and ten seeds. Old plans are exact cuts; handoffs cost one. All cases remain (generators/freeze history: Supplement S4).

Per-case medians of three alternating timings precede paired ratios. Intervals resample $(n, \mathrm{family}, \mathrm{seed})$ clusters 2,000 times with factors/repeats together; the whole grid was not prospectively registered. N denotes queries not resolved by the shared cut shortcut.

All 2,880 calls prove equal optima: 318/480 queries are not shortcut-resolved, while 314 budgets strictly increase the

TABLE II
MAIN STUDY: MODEL SIZE, ABSOLUTE TIME AND PAIRED SPEED.

| Family, $n$ | N/all | $K/M\%$ | F/R ms | Speed N/all |
|---|---|---|---|---|
| Grid, 128 | 69/80 | 98.4/98.3 | 34.5/30.4 | 1.00/1.00 |
| Grid, 512 | 54/80 | 53.7/52.4 | 131.0/64.6 | 1.88/1.37 |
| Modular, 128 | 29/80 | 71.9/70.3 | 23.2/19.6 | 1.30/1.00 |
| Modular, 512 | 30/80 | 25.9/24.8 | 74.9/23.3 | 3.17/1.01 |
| Small-world, 128 | 72/80 | 100.0/100.0 | 70.7/72.8 | 0.98/0.99 |
| Small-world, 512 | 64/80 | 100.0/100.0 | 354.7/355.6 | 0.99/0.99 |

TABLE III
WEIGHTED ABLATION. REGION COLUMNS: NON-SHORTCUT MEDIAN PERCENTAGES. SPEED: HOPCP/BARRCP.

| Family | N/all | $H_B\%$ | $K_B\%$ | Speed N | Speed all |
|---|---|---|---|---|---|
| Grid | 56/80 | 56.7 | 20.0 | 2.59 | 2.13 |
| Modular | 40/80 | 22.0 | 9.8 | 1.56 | 1.03 |
| Small-world | 66/80 | 100.0 | 99.5 | 1.01 | 1.01 |

optimum (a rejected cut can have a feasible tie). Greedy is suboptimal on 55/480. Table II asks: how often does the shortcut fail, how much model remains, and does search saved exceed preprocessing? At 512 modular vertices, only 30/80 are non-shortcut; median retained labels/constructed Booleans are 25.9/24.8%. Full/BARR medians are 74.9/23.3 ms, with 1.72 ms BARR preprocessing. Paired speed is 3.17, interval $[2.86, 3.86]$, but 1.01 overall. Grid speeds are 1.88, $[1.54, 2.15]$, and 1.37. Supplementary Fig. S1 shows geometry–speed associations, not a single-factor causal effect.

Table II gives N-query median retained labels/Booleans ($K/M$, %), $M = (|K|+|E[K]|)/(n+m)$ before presolve, and marginal Full/BARR time medians. Speed uses paired ratios, N/all. Over all 80 modular-512 cases, sums of per-case median times are 3.634/1.540 s (ratio 2.36); all strata/components appear in Supplement S1–S2.

A separate 240-combination, 512-vertex ablation uses $r_i$ uniform in $\{1, 2, 3\}$ and seeds 200–209. FullCP, safe-hop HopCP, and BARRCP rotate through three timing positions; both regional methods pay for certification. All 2,160 calls prove the same global optima. Table III shows that weighted geometry helps grids and modules but scarcely reduces small-world graphs. Grid HopCP/BARRCP speed has interval $[2.41, 2.75]$, modular $[1.47, 1.68]$; repetitions are not independent cases.

For 32 inexact old plans on 128-vertex grids/modules, one/three low-cost non-anchor flips give old gaps 24–105 (median 58), including interactions. Both full and restricted optima are proved. For $\Delta = \mathrm{OPT}_K - \mathrm{OPT}_B$, 16/26 genuinely reduced cases have zero loss; six retain the full graph and necessarily lose nothing (22/32 zero overall). Ten have positive loss (maximum 67); three attain $\Delta = \varepsilon$. The ratio $\Delta/\varepsilon$ has median 0, IQR $[0, 0.334]$ and range $[0, 1]$ (Supplementary Fig. S2). This is descriptive tightness, not acceptability or self-certification: zero loss uses the proved full reference. Unequal-quality outputs are excluded from speed comparisons. Checks include 24 tests and 1,200 earlier enumerations.

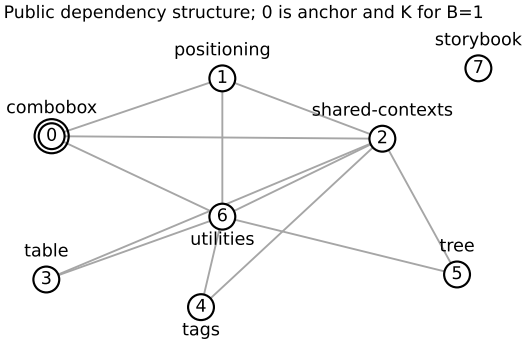



Fig. 1. Eight packages with publicly listed alternative owners [2]; numbers index packages. Edges are runtime dependencies with unit proxy cost. The double circle marks the synthetic anchor and retained region for $B = 1$. Routing labels and cost changes are illustrative.

### B. Source-Grounded Responsibility Case

Fluent UI commit `941ed35` [2] lists exactly teams `cxe-prg` and `teams-prg` for eight package scopes; we map 0/1 to these teams globally. Their manifests induce 12 runtime-dependency edges, retaining isolates (Fig. 1). Build/configuration exceptions and outside packages are excluded. We propose primary-contact routing over these alternatives, without changing permissions or modeling review queues. Declarations are observed; live access, expertise, primary routing, costs and reserved effort are unverified.

For a transparent worked repair, sorted packages alternate synthetic preferred teams, with costs 0/1; unit edges and handoffs define a synthetic old cut optimum. Its labels are $(0, 0, 0, 0, 0, 0, 0, 1)$ and old value is 3. Increasing combobox's old-label cost by 12 makes an unconstrained optimum change seven packages, exceeding $B = 1$. BARR keeps only combobox and attains the enumerated full budgeted optimum 7, versus 15 for no repair. For unit loads, old loads are $(7, 1)$; $W(B) = (1, 1)$ certifies hypothetical capacities $(8, 2)$. Capacities $(7, 2)$ fail the check: no reduced capacity guarantee is returned.

All eight anchors, increases 4/12 and budgets 1–4 yield 64 repairs. Both guarded CP routes match complete enumeration; 40 budgets strictly raise the optimum and eight use the shortcut. This is exploratory: the retained cost-4 diagnostic had 64/64 shortcuts. Synthetic costs/reservations and this small portfolio establish neither runtime gains nor operational event frequency.

### C. Public CAD Structure Audit

The Bath logs [4], [5] record three vehicle-design projects, not interchangeable teams. Retaining files with at least five distinct edit times yields 226/231/168 nodes. Each edit counts other files once until the next same-file edit or $h = 1, 12$ hours; averaged directional frequencies are thresholded at $1/4$ and rounded after scaling by eight. Isolates remain. Edge counts are 2726/8152, 535/693 and 199/481: co-edit proxies, not physical interfaces (Fig. 2). Synthetic preferences/costs,

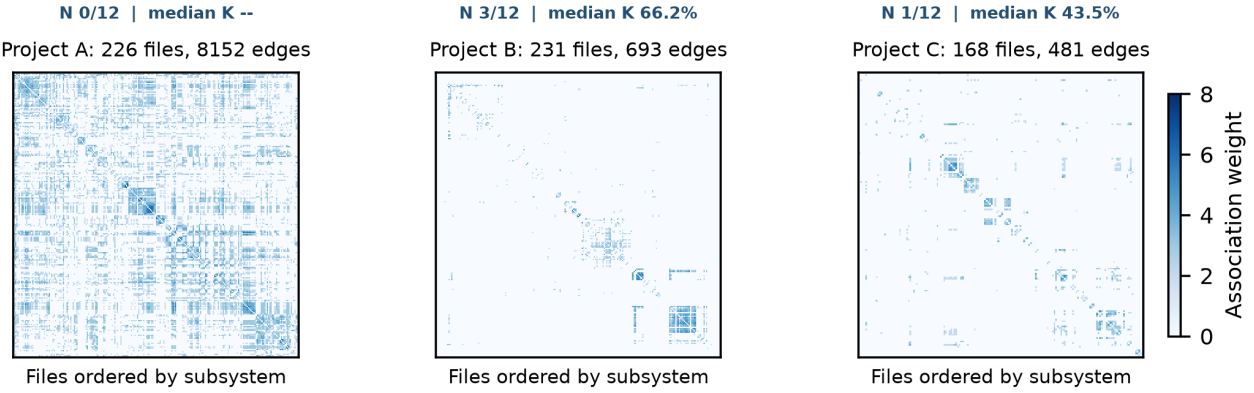



Fig. 2. Twelve-hour CAD co-edit matrices from public logs [5], ordered by subsystem/file ID. Top labels show N-query counts and their median retained vertex fraction, matching Table IV. Color is an association proxy, not measured handoff effort or physical interfaces. Isolates remain; ownership/costs are synthetic.

one/four anchors, $B = 8, 16$ and three seeds define 72 cases frozen before outcomes (full generator supplied). All CP pairs prove equal values, but 66 use the shared shortcut (14 without changes). Qualification and slack are unobserved. This shows structure compatibility, not operational acceleration or how frequently reduction helps.

### D. Lagrangian Route and Larger/Harder Inputs

LRRepair is a classical comparator, not a named optimizer reproduction. Rational cuts of $E_1+\lambda R$ give bounds $q(\lambda)-\lambda B$, rounded upward for integer objectives. Budget restoration and feasible single-flip descent supply incumbents, with the old plan as fallback. Dual search gets at most 65% of the target, recovery the rest (details in Supplement S4). Finite search need not optimize the dual; recovery has no approximation guarantee.

The three-route stage reuses all 120 large and 20 core inputs plus 72 public cases (636 calls). Scale inputs cross $n = 1024, 2048$, three families, heterogeneous costs and five seeds, frozen after 32 exploratory cases (full factors: Supplement S4). Cores of $q = 64, 128$ plus dominated tails separate search difficulty from preprocessing size; old plans are provably optimal.

Targets are three seconds for scale/public and five for cores, with rotating method order and one timing each. These descriptive comparisons reuse earlier inputs; they are not additional independent instances or pooled with triple-timing intervals. Table IV retains every censored call and reports quality alongside time.

At 2,048 vertices, 14/20 modular cases are non-shortcut; their median paired speed is 7.30, with all-case medians 0.148/0.016 s. On 64-cores at that size both CP routes certify 5/5, with paired speed 3.24. LRRepair's median is 0.173 s but maximum actual loss is 12 against proved optima, with no certificates. Its maximum loss is 26 on eight proved 128-core references. BARRCP certifies 4/5 per size in this single stage; Table V checks repeated queries, not a stable success probability. Earlier stages remain in Supplement S4.

LRRepair matches 71/72 public optima, certifies 69/72 and loses at most one. It matches every proved grid optimum, with some dual bounds open. On four censored 2,048-vertex small-world cases it improves both CP incumbents by up to 417; its maximum gap is 8 versus 418 for CP, and reduction remains ineffective. Cut recovery can favor fast feasible repair, BARR exact certification on suitable structures; neither dominates time, quality and bound closure.

### E. Repeatability of Difficult Core Queries

All 20 saved cores have ten repeats per CP route (400 calls): five-second soft target, one worker, fixed solver seeds, shuffled cases and alternating methods. Continuous monotonic time and clock/pause checks (0.1 s threshold) pass for all calls in two execution segments; saved calls are not rerun. No concurrent benchmark/build was launched. Separate suspend diagnostics and interruption history are in Supplement S4.

Table V retains all incomplete queries. Small dispersion at the cap reflects censoring, not reliable time-to-optimum. Uncontrolled background load/power and fixed-seed single-laptop repeats preclude cross-session reliability claims. Closure uses each call's own interval. We give no unresolved-core speed ratio or independent-trial interval; all per-case outcomes and timing components are supplied.

## VI. Discussion and Limitations

Useful speed needs a rejected cut, a smaller retained model and search savings beyond preprocessing. Dense graphs can retain everything; the backend may remain exponential. Each new event needs recertification: a previous budgeted optimum need not be old-unconstrained optimal. Ownership roles are sourced, but costs/capacities are hypothetical; the guard needs real loads/reservations. CAD co-edits establish neither eligibility nor human efficiency. Active capacities, scheduling and multiple teams need new feasibility arguments. No industrial trial or peak-memory study is provided; single-laptop timings/repeats do not establish cross-machine stability. Sequential repair and reversion-closed feasible sets remain open.

TABLE IV

THREE-ROUTE COMPARISON: F/R/L = FULLCP/BARRCP/LRREPAIR. TIME: ALL-CASE MEDIAN SECONDS; CERT: CLOSED BOUNDS; GAP: MAXIMUM RETURNED ABSOLUTE BOUND GAP, INCLUDING INCOMPLETE CALLS. $K$: NON-SHORTCUT MEDIAN REGION PERCENTAGE (–: ALL SHORTCUT). FASTER INCUMBENTS NEED NOT HAVE EQUAL QUALITY.

| Input | $n$ | N/all | $K\%$ | Cert F/R/L | Time F/R/L (s) | Max gap F/R/L |
|---|---|---|---|---|---|---|
| Grid | 1024 | 20/20 | 70.5 | 20/20/11 | 0.189/0.115/0.081 | 0/0/7 |
| Grid | 2048 | 16/20 | 50.4 | 20/20/15 | 0.289/0.067/0.187 | 0/0/3 |
| Modular | 1024 | 15/20 | 13.7 | 20/20/6 | 0.075/0.012/0.059 | 0/0/26 |
| Modular | 2048 | 14/20 | 7.1 | 20/20/6 | 0.148/0.016/0.126 | 0/0/24 |
| Small-world | 1024 | 20/20 | 100.0 | 20/20/6 | 0.256/0.268/0.143 | 0/0/17 |
| Small-world | 2048 | 20/20 | 100.0 | 16/16/10 | 0.639/0.627/0.320 | 418/418/8 |
| Core 64 | 1024 | 5/5 | 9.4 | 5/5/0 | 0.440/0.215/0.082 | 0/0/21 |
| Core 64 | 2048 | 5/5 | 4.7 | 5/5/0 | 0.641/0.221/0.173 | 0/0/21 |
| Core 128 | 1024 | 5/5 | 18.8 | 2/4/0 | 5.002/3.548/0.120 | 15/17/43 |
| Core 128 | 2048 | 5/5 | 9.4 | 1/4/0 | 5.004/3.756/0.235 | 20/17/43 |
| CAD A, 1 h | 226 | 0/12 | – | 12/12/12 | 0.0013/0.0013/0.0018 | 0/0/0 |
| CAD A, 12 h | 226 | 0/12 | – | 12/12/12 | 0.0036/0.0037/0.0051 | 0/0/0 |
| CAD B, 1 h | 231 | 2/12 | 24.9 | 12/12/10 | 0.0004/0.0004/0.0005 | 0/0/15 |
| CAD B, 12 h | 231 | 3/12 | 66.2 | 12/12/11 | 0.0004/0.0004/0.0006 | 0/0/1 |
| CAD C, 1 h | 168 | 0/12 | – | 12/12/12 | 0.0002/0.0002/0.0003 | 0/0/0 |
| CAD C, 12 h | 168 | 1/12 | 43.5 | 12/12/12 | 0.0003/0.0003/0.0005 | 0/0/0 |

TABLE V

FIVE FIXED INPUTS/ROW, TEN REPEATS/METHOD (F/R = FULLCP/BARRCP). CLOSED COUNTS: /50; RANGES: PER-INPUT /10. TIMES: ALL-QUERY MEDIAN [Q1,Q3], INCLUDING INCOMPLETE CALLS, NOT UNCENSORED TIME-TO-OPTIMUM. REL. IQR: MEDIAN ACROSS INPUTS OF (Q3–Q1)/MEDIAN TIME (%). REPEATS ARE NOT 50 INDEPENDENT INPUTS.

| $q$ | $n$ | Closed F/R | Per-case F/R | Query time F / R (s) | Rel. IQR F/R | Max gap F/R |
|---|---|---|---|---|---|---|
| 64 | 1024 | 50/50 | 10–10/10–10 | 0.453 [0.311,0.604] / 0.216 [0.211,0.238] | 2.4/1.9 | 0/0 |
| 64 | 2048 | 50/50 | 10–10/10–10 | 0.615 [0.545,0.783] / 0.218 [0.214,0.242] | 1.5/4.0 | 0/0 |
| 128 | 1024 | 18/40 | 0–9/0–10 | 5.002 [4.393,5.002] / 3.517 [3.018,4.073] | 0.0/2.6 | 17/17 |
| 128 | 2048 | 15/39 | 0–9/0–10 | 5.004 [4.945,5.004] / 3.548 [3.006,3.922] | 0.0/1.3 | 20/17 |

## VII. CONCLUSION

Union reversion yields certified region reduction for localized binary responsibility repair, with conditional modular/grid benefits. Public ownership records ground a proposed use, and a capacity check tests static slack. Operational gains and active-capacity extensions remain unestablished.

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